QCar2 Internship

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Flatness-Based Bézier Trajectory Generation for QCar2

1 Introduction

For trajectory planning, the QCar2 is approximated by a car-like kinematic model, which provides a simpler flatness-based representation than the bicycle dynamic model. Let x and y denote the Cartesian coordinates of the midpoint of the rear axle in the inertial frame, θ the vehicle heading angle, and φ the front-wheel steering angle. Let v be the longitudinal velocity, ω^s the steering angular velocity, and l the wheelbase of the vehicle. Moreover, the Cartesian coordinates of the midpoint of the front axle are given by

$$x_f = x + l\cos\theta, \qquad y_f = y + l\sin\theta. \tag{1}$$

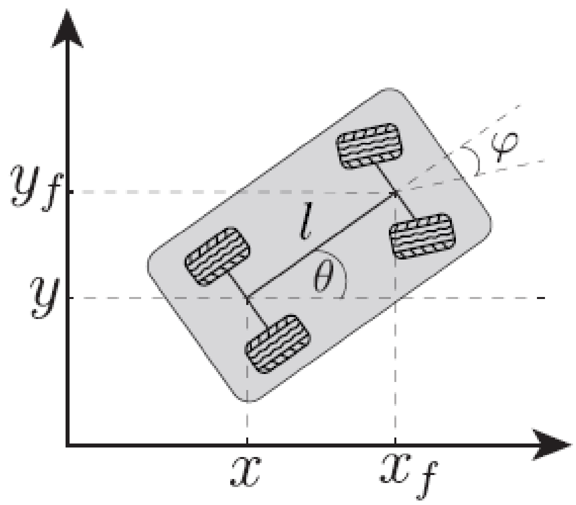

Figure 1: Car-like vehicle model.

The state and input vectors are defined as

$$\mathbf{x}_k = \begin{bmatrix} x & y & \theta & \varphi \end{bmatrix}^\top, \qquad \mathbf{u}_k = \begin{bmatrix} v & \omega^s \end{bmatrix}^\top.$$

Under the pure-rolling and no-slip assumptions, the car-like model is given by

$$\dot{x} = v\cos\theta \tag{2a}$$
$$\dot{y} = v\sin\theta \tag{2b}$$
$$\dot{\theta} = \frac{v}{l}\tan\varphi \tag{2c}$$
$$\dot{\varphi} = \omega^s. \tag{2d}$$

2 Differential Flatness

2.1 State and Input Representation in Terms of the Flat Output

For the car-like kinematic model, the flat output is chosen as the Cartesian position of the midpoint of the rear axle:

$$\mathbf{z} = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}. \tag{3}$$

Since

$$\dot{x} = v\cos\theta, \qquad \dot{y} = v\sin\theta,$$

The first derivative of the flat output is

$$\dot{\mathbf{z}} = \begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \end{bmatrix} = \begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} v\cos\theta \\ v\sin\theta \end{bmatrix}.$$

Therefore, the longitudinal velocity and the vehicle heading angle can be directly recovered from the flat output as

$$v = \sqrt{\dot{z}_1^2 + \dot{z}_2^2}, \qquad \theta = \arctan\left(\frac{\dot{z}_2}{\dot{z}_1}\right). \tag{4}$$

To recover the steering angle, we differentiate the heading angle. From

$$\theta = \arctan\left(\frac{\dot{z}_2}{\dot{z}_1}\right),$$

Its time derivative is

$$\dot{\theta} = \frac{\dot{z}_1\ddot{z}_2 - \dot{z}_2\ddot{z}_1}{\dot{z}_1^2 + \dot{z}_2^2}.$$

From the car-like model,

$$\dot{\theta} = \frac{v}{l}\tan\varphi.$$

we obtain

$$\tan\varphi = \frac{l\left(\dot{z}_1\ddot{z}_2 - \dot{z}_2\ddot{z}_1\right)}{\left(\dot{z}_1^2 + \dot{z}_2^2\right)^{3/2}}. \tag{5}$$

Thus, the steering angle is given by

$$\varphi = \arctan\left(\frac{l\left(\dot{z}_1\ddot{z}_2 - \dot{z}_2\ddot{z}_1\right)}{\left(\dot{z}_1^2 + \dot{z}_2^2\right)^{3/2}}\right). \tag{6}$$

Finally, the steering angular velocity is obtained by differentiating the steering angle:

$$\omega^s = \dot{\varphi}.$$

Let

$$A = \dot{z}_1\ddot{z}_2 - \dot{z}_2\ddot{z}_1, \qquad B = \left(\dot{z}_1^2 + \dot{z}_2^2\right)^{3/2}.$$

Then

$$\varphi = \arctan\left(\frac{lA}{B}\right).$$

The derivative of A is

$$\dot{A} = \dot{z}_1 z_2^{(3)} - \dot{z}_2 z_1^{(3)}.$$

The derivative of B is

$$\dot{B} = 3\left(\dot{z}_1^2 + \dot{z}_2^2\right)^{1/2}\left(\dot{z}_1\ddot{z}_1 + \dot{z}_2\ddot{z}_2\right).$$

Therefore, the steering angular velocity can be written as

$$\omega^s = \frac{l\left(\dot{A}B - A\dot{B}\right)}{B^2 + l^2A^2}.$$

Substituting A, $\dot{A}$, B, and $\dot{B}$ into the above expression gives

$$\omega^s = \frac{l\left[\left(\dot{z}_1 z_2^{(3)} - \dot{z}_2 z_1^{(3)}\right)\left(\dot{z}_1^2 + \dot{z}_2^2\right)^{3/2} - 3\left(\dot{z}_1\ddot{z}_2 - \dot{z}_2\ddot{z}_1\right)\left(\dot{z}_1^2 + \dot{z}_2^2\right)^{1/2}\left(\dot{z}_1\ddot{z}_1 + \dot{z}_2\ddot{z}_2\right)\right]}{\left(\dot{z}_1^2 + \dot{z}_2^2\right)^3 + l^2\left(\dot{z}_1\ddot{z}_2 - \dot{z}_2\ddot{z}_1\right)^2}. \tag{7}$$

Consequently, all the states and inputs of the car-like model can be expressed as functions of the flat output $\mathbf{z}$ and its derivatives:

$$x = z_1, \qquad y = z_2, \qquad \theta = \arctan\left(\frac{\dot{z}_2}{\dot{z}_1}\right), \qquad \varphi = \arctan\left(\frac{lA}{B}\right),$$

and

$$v = \sqrt{\dot{z}_1^2 + \dot{z}_2^2}, \qquad \omega^s = \frac{l\left(\dot{A}B - A\dot{B}\right)}{B^2 + l^2 A^2}.$$

2.2 Flat Output Derivatives in Terms of States and Inputs

The boundary conditions can be written in terms of the vehicle states and inputs. At a given time instant, let

$$\mathbf{x} = \begin{bmatrix} x & y & \theta & \varphi \end{bmatrix}^{\mathrm{T}}, \qquad \mathbf{u} = \begin{bmatrix} v & \omega^s \end{bmatrix}^{\mathrm{T}}.$$

Using the flatness relations, the corresponding flat output and its derivatives are expressed as

$$\mathbf{z} = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}. \tag{8}$$

From the kinematic model, the first derivative is

$$\dot{\mathbf{z}} = \begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \end{bmatrix} = \begin{bmatrix} v\cos\theta \\ v\sin\theta \end{bmatrix}. \tag{9}$$

The second derivative is obtained by differentiating $\dot{\mathbf{z}}$:

$$\ddot{z}_1 = \dot{v}\cos\theta - v\dot{\theta}\sin\theta,$$

$$\ddot{z}_2 = \dot{v}\sin\theta + v\dot{\theta}\cos\theta.$$

Since

$$\dot{\theta} = \frac{v}{l}\tan\varphi,$$

we obtain

$$\ddot{\mathbf{z}} = \begin{bmatrix} \dot{v}\cos\theta - \dfrac{v^2}{l}\tan\varphi\sin\theta \\ \dot{v}\sin\theta + \dfrac{v^2}{l}\tan\varphi\cos\theta \end{bmatrix}. \tag{10}$$

The third derivative is then obtained by differentiating $\ddot{\mathbf{z}}$. For compactness, define

$$\Omega = \dot{\theta} = \frac{v}{l}\tan\varphi, \qquad \dot{\Omega} = \frac{\dot{v}}{l}\tan\varphi + \frac{v}{l}\sec^2\varphi\,\omega^s, \qquad where\sec^2\varphi = \frac{1}{\cos^2\varphi}.$$

Then,

$$\mathbf{z}^{(3)} = \begin{bmatrix} \ddot{v}\cos\theta - 2\dot{v}\Omega\sin\theta - v\dot{\Omega}\sin\theta - v\Omega^2\cos\theta \\ \ddot{v}\sin\theta + 2\dot{v}\Omega\cos\theta + v\dot{\Omega}\cos\theta - v\Omega^2\sin\theta \end{bmatrix}. \tag{11}$$

At the boundary conditions, the longitudinal velocity is assumed to be constant, i.e.,

$$v = \text{const} \quad \Rightarrow \quad \dot{v} = 0, \quad \ddot{v} = 0.$$

Under this assumption, the second derivative of the flat output becomes

$$\ddot{\mathbf{z}} = \begin{bmatrix} -\dfrac{v^2}{l}\tan\varphi\sin\theta \\ \dfrac{v^2}{l}\tan\varphi\cos\theta \end{bmatrix}. \tag{12}$$

Similarly, using
$$\dot{\theta} = \Omega = \frac{v}{l}\tan\varphi, \qquad \dot{\Omega} = \frac{v}{l}\sec^2\varphi\,\omega^s,$$
The third derivative simplifies to
$$\mathbf{z}^{(3)} = \begin{bmatrix} -v\dot{\Omega}\sin\theta - v\Omega^2\cos\theta \\ v\dot{\Omega}\cos\theta - v\Omega^2\sin\theta \end{bmatrix}. \tag{13}$$

3 Trajectory Generating

3.1 Flatness-Based Bézier Trajectory Generating

The trajectory is generated from the flat output
$$\mathbf{z}(t) = \begin{bmatrix} z_1(t) \\ z_2(t) \end{bmatrix} = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}.$$

In this work, the boundary conditions are imposed on the flat output and its derivatives up to the third order at both the initial and final times:
$$\mathbf{z}(0), \quad \dot{\mathbf{z}}(0), \quad \ddot{\mathbf{z}}(0), \quad \mathbf{z}^{(3)}(0),$$
$$\mathbf{z}(T), \quad \dot{\mathbf{z}}(T), \quad \ddot{\mathbf{z}}(T), \quad \mathbf{z}^{(3)}(T).$$

For a Bézier polynomial of degree n, there are $n+1$ control points. Since the boundary conditions are imposed at both the initial and final times, the total number of boundary constraints is $2(r+1)$, where r is the highest derivative order. In this work, the highest derivative order is $r = 3$. Therefore,
$$n + 1 = 2(3+1) = 8 => n = 7$$

The Bézier polynomial of degree 7 is defined as
$$\mathbf{z}(s) = \sum_{i=0}^{7} \mathbf{P}_i B_i^7(s), \qquad s \in [0,1], \tag{14}$$
where
$$B_i^7(s) = \binom{7}{i}(1-s)^{7-i}s^i, \qquad i = 0, 1, \ldots, 7. \tag{15}$$

Since the Bézier parameter is defined over the interval $s \in [0,1]$, while the physical time evolves over $t \in [0,T]$, where T denotes the total trajectory duration, we introduce the normalized time variable
$$s = \frac{t}{T}.$$

Thus, using the chain rule, the time derivative can be written as
$$\frac{ds}{dt} = \frac{1}{T}.$$

First derivative of the Bézier polynomial
Starting from the Bézier polynomial,
$$\mathbf{z}(s) = \sum_{i=0}^{7} \mathbf{P}_i B_i^7(s),$$
where
$$B_i^7(s) = \binom{7}{i}(1-s)^{7-i}s^i.$$

Differentiating with respect to s, we obtain
$$\frac{d\mathbf{z}}{ds} = \sum_{i=0}^{7} \mathbf{P}_i \frac{d}{ds}\left[\binom{7}{i}(1-s)^{7-i}s^i\right].$$

Using the product rule,

$$\begin{aligned}\frac{dB_i^7(s)}{ds} &= \frac{d}{ds}\left[\binom{7}{i}(1-s)^{7-i}s^i\right] \\ &= \binom{7}{i}\left[-(7-i)(1-s)^{6-i}s^i + i(1-s)^{7-i}s^{i-1}\right].\end{aligned}$$

Expanding the derivative of the Bézier polynomial gives

$$\begin{aligned}\frac{d\mathbf{z}}{ds} &= \sum_{i=0}^{7}\mathbf{P}_i\frac{dB_i^7(s)}{ds} \\ &= \sum_{i=0}^{7}\mathbf{P}_i\binom{7}{i}\left[-(7-i)(1-s)^{6-i}s^i + i(1-s)^{7-i}s^{i-1}\right] \\ &= \sum_{i=0}^{7}\left[-\mathbf{P}_i\binom{7}{i}(7-i)(1-s)^{6-i}s^i + \mathbf{P}_i\binom{7}{i}i(1-s)^{7-i}s^{i-1}\right].\end{aligned}$$

Using

$$\binom{7}{i}(7-i) = 7\binom{6}{i}, \qquad \binom{7}{i}i = 7\binom{6}{i-1},$$

we obtain

$$\begin{aligned}\frac{d\mathbf{z}}{ds} &= \sum_{i=0}^{7}\left[-7\mathbf{P}_i\binom{6}{i}(1-s)^{6-i}s^i + 7\mathbf{P}_i\binom{6}{i-1}(1-s)^{7-i}s^{i-1}\right] \\ &= 7\sum_{i=0}^{7}\mathbf{P}_i\left[B_{i-1}^6(s) - B_i^6(s)\right].\end{aligned}$$

Expanding the summation gives

$$\begin{aligned}\frac{d\mathbf{z}}{ds} = 7\big[&\mathbf{P}_0(B_{-1}^6 - B_0^6) + \mathbf{P}_1(B_0^6 - B_1^6) + \cdots \\ &+ \mathbf{P}_6(B_5^6 - B_6^6) + \mathbf{P}_7(B_6^6 - B_7^6)\big].\end{aligned}$$

Since $B_{-1}^6(s) = 0$ and $B_7^6(s) = 0$, we have

$$\begin{aligned}\frac{d\mathbf{z}}{ds} = 7\big[&(\mathbf{P}_1 - \mathbf{P}_0)B_0^6(s) + (\mathbf{P}_2 - \mathbf{P}_1)B_1^6(s) + \cdots \\ &+ (\mathbf{P}_7 - \mathbf{P}_6)B_6^6(s)\big].\end{aligned}$$

Thus, the first derivative with respect to s is

$$\frac{d\mathbf{z}}{ds} = 7\sum_{i=0}^{6}\left(\mathbf{P}_{i+1} - \mathbf{P}_i\right)B_i^6(s).$$

Since $s = t/T$, the first time derivative is

$$\dot{\mathbf{z}}(t) = \frac{7}{T}\sum_{i=0}^{6}\left(\mathbf{P}_{i+1} - \mathbf{P}_i\right)B_i^6(s). \tag{16}$$

The second time derivative is

$$\ddot{\mathbf{z}}(t) = \frac{42}{T^2}\sum_{i=0}^{5}\left(\mathbf{P}_{i+2} - 2\mathbf{P}_{i+1} + \mathbf{P}_i\right)B_i^5(s). \tag{17}$$

The third time derivative is

$$\mathbf{z}^{(3)}(t) = \frac{210}{T^3}\sum_{i=0}^{4}\left(\mathbf{P}_{i+3} - 3\mathbf{P}_{i+2} + 3\mathbf{P}_{i+1} - \mathbf{P}_i\right)B_i^4(s). \tag{18}$$

At $s = 0$, the first four control points are obtained from the initial boundary conditions:

$$\mathbf{P}_0 = \mathbf{z}(0), \tag{19}$$

$$\mathbf{P}_1 = \mathbf{z}(0) + \frac{T}{7}\dot{\mathbf{z}}(0), \tag{20}$$

$$\mathbf{P}_2 = \mathbf{z}(0) + \frac{2T}{7}\dot{\mathbf{z}}(0) + \frac{T^2}{42}\ddot{\mathbf{z}}(0), \tag{21}$$

$$\mathbf{P}_3 = \mathbf{z}(0) + \frac{3T}{7}\dot{\mathbf{z}}(0) + \frac{T^2}{14}\ddot{\mathbf{z}}(0) + \frac{T^3}{210}\mathbf{z}^{(3)}(0). \tag{22}$$

At $s = 1$, the remaining control points are obtained from the final boundary conditions:

$$\mathbf{P}_7 = \mathbf{z}(T), \tag{23}$$

$$\mathbf{P}_6 = \mathbf{z}(T) - \frac{T}{7}\dot{\mathbf{z}}(T), \tag{24}$$

$$\mathbf{P}_5 = \mathbf{z}(T) - \frac{2T}{7}\dot{\mathbf{z}}(T) + \frac{T^2}{42}\ddot{\mathbf{z}}(T), \tag{25}$$

$$\mathbf{P}_4 = \mathbf{z}(T) - \frac{3T}{7}\dot{\mathbf{z}}(T) + \frac{T^2}{14}\ddot{\mathbf{z}}(T) - \frac{T^3}{210}\mathbf{z}^{(3)}(T). \tag{26}$$

3.2 Automatic Computation of Boundary Conditions from Waypoint Geometry

In practice, the user specifies the Cartesian position of all waypoints, the heading angle at the first and last waypoints, and the travelling time of each segment. For a sequence of N waypoints, the input data are defined as

$$\mathbf{p}_i = \begin{bmatrix} x_i \\ y_i \end{bmatrix}, \qquad i = 0, \ldots, N-1,$$

together with the prescribed boundary headings

$$\theta_0, \qquad \theta_{N-1},$$

and the segment times

$$T_i, \qquad i = 0, \ldots, N-2.$$

The remaining quantities required by the flatness-based trajectory generation, namely, the intermediate heading angles, the longitudinal velocities, the steering angles, and the steering angular velocities, are automatically computed from the waypoint geometry and the segment times.

For an intermediate waypoint i, **the local heading direction** is estimated from the incoming and outgoing geometric directions. The normalized incoming and outgoing direction vectors are

$$\mathbf{d}_i^- = \frac{\mathbf{p}_i - \mathbf{p}_{i-1}}{\|\mathbf{p}_i - \mathbf{p}_{i-1}\|}, \qquad \mathbf{d}_i^+ = \frac{\mathbf{p}_{i+1} - \mathbf{p}_i}{\|\mathbf{p}_{i+1} - \mathbf{p}_i\|}.$$

The local tangent direction is then approximated by

$$\tilde{\mathbf{h}}_i = \mathbf{d}_i^- + \mathbf{d}_i^+.$$

The corresponding unit heading direction is obtained by normalization:

$$\hat{\mathbf{h}}_i = \frac{\tilde{\mathbf{h}}_i}{\|\tilde{\mathbf{h}}_i\|} = \begin{bmatrix} \cos\theta_i \\ \sin\theta_i \end{bmatrix}.$$

Therefore, the heading angle at the intermediate waypoint is obtained as

$$\theta_i = \operatorname{atan2}\left(\hat{\mathbf{h}}_{i,y}, \hat{\mathbf{h}}_{i,x}\right), \qquad i = 1, \ldots, N-2.$$

The heading angles at the first and last waypoints are not computed from the geometry, but are directly imposed by the user.

The path curvature at an intermediate waypoint is required in order to compute the steering angle. From the car-like kinematic model, the heading dynamics are given by

$$\dot{\theta} = \frac{v}{l} \tan \varphi.$$

For a planar path, the curvature is defined as the variation of the heading angle with respect to the travelled distance:

$$\kappa = \frac{d\theta}{ds}.$$

Since the travelled distance satisfies $ds/dt = v$, the curvature can also be written as

$$\kappa = \frac{d\theta/dt}{ds/dt} = \frac{\dot{\theta}}{v}.$$

Substituting the car-like heading dynamics into this expression gives

$$\kappa = \frac{1}{v}\left(\frac{v}{l} \tan \varphi\right) = \frac{\tan \varphi}{l}.$$

Therefore, if the path curvature is known, the steering angle can be computed as

$$\varphi = \arctan(l\kappa).$$

However, at this stage, the curvature cannot be computed directly from $\kappa = \dot{\theta}/v$. Although the heading angles are known at the waypoints, they are only discrete boundary orientations. They do not yet define a continuous and consistently unwrapped heading profile $\theta(t)$ along the whole path. Moreover, the first and last headings are imposed by the user and may not exactly coincide with the local geometric tangent of the waypoint sequence.

For this reason, the curvature used for the steering-angle computation is estimated directly from the geometry of three consecutive waypoints $\mathbf{p}_{i-1}$, $\mathbf{p}_i$, and $\mathbf{p}_{i+1}$. This provides a purely geometric approximation of the local path bending before the continuous Bézier trajectory is constructed.

The three vectors involved in the curvature computation are

$$\mathbf{p}_i - \mathbf{p}_{i-1} = \begin{bmatrix} x_i - x_{i-1} \\ y_i - y_{i-1} \end{bmatrix}, \qquad \mathbf{p}_{i+1} - \mathbf{p}_i = \begin{bmatrix} x_{i+1} - x_i \\ y_{i+1} - y_i \end{bmatrix}, \qquad \mathbf{p}_{i+1} - \mathbf{p}_{i-1} = \begin{bmatrix} x_{i+1} - x_{i-1} \\ y_{i+1} - y_{i-1} \end{bmatrix}.$$

The signed curvature at an intermediate waypoint is then approximated as

$$\kappa_i = \frac{2\left[(x_i - x_{i-1})(y_{i+1} - y_i) - (y_i - y_{i-1})(x_{i+1} - x_i)\right]}{\sqrt{(x_i - x_{i-1})^2 + (y_i - y_{i-1})^2}\sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2}\sqrt{(x_{i+1} - x_{i-1})^2 + (y_{i+1} - y_{i-1})^2}}.$$

The numerator corresponds to the signed 2D cross product between two consecutive path segments. Hence, its sign indicates the turning direction of the path. A positive value corresponds to a left turn, while a negative value corresponds to a right turn. If the three points are aligned, the numerator is zero and the curvature is also zero.

At the first and last waypoints, the curvature is set to zero:

$$\kappa_0 = 0, \qquad \kappa_{N-1} = 0.$$

Finally, the steering angle at each waypoint is obtained from the geometric curvature as

$$\varphi_i = \arctan(l\kappa_i).$$

The longitudinal velocity is then computed from the segment times. The length of segment i, connecting waypoint i to waypoint $i+1$, is

$$d_i = \|\mathbf{p}_{i+1} - \mathbf{p}_i\|.$$

Since the segment time T_i is prescribed, the average velocity along this segment is

$$\bar{v}_i = \frac{d_i}{T_i}.$$

However, the flatness-based boundary conditions require a longitudinal velocity at each waypoint. Therefore, the waypoint velocities are obtained from the adjacent segment velocities. At the first waypoint,

$$v_0 = \bar{v}_0.$$

At the last waypoint,

$$v_{N-1} = \bar{v}_{N-2}.$$

For an intermediate waypoint, the velocity is chosen as the average of the two neighboring segment velocities:

$$v_i = \frac{\bar{v}_{i-1} + \bar{v}_i}{2}, \qquad i = 1, \dots, N-2.$$

After the steering angles and the segment times are known, **the steering angular velocity** is approximated from the variation of the steering angle. For the first waypoint,

$$\omega_0^s = \frac{\varphi_1 - \varphi_0}{T_0}.$$

For the last waypoint,

$$\omega_{N-1}^s = \frac{\varphi_{N-1} - \varphi_{N-2}}{T_{N-2}}.$$

For an intermediate waypoint, a centered approximation is used:

$$\omega_i^s = \frac{\varphi_{i+1} - \varphi_{i-1}}{T_{i-1} + T_i}, \qquad i = 1, \dots, N-2.$$

Thus, the initial user-defined data

$$\mathbf{p}_i, \qquad \theta_0, \qquad \theta_{N-1}, \qquad T_i$$

are converted into complete boundary waypoints of the form

$$\mathbf{w}_i = \begin{bmatrix} x_i & y_i & \theta_i & \varphi_i & v_i & \omega_i^s \end{bmatrix}^\top.$$

These complete boundary waypoints are then used to construct the flat output boundary conditions required by the degree-7 Bézier trajectory generation. Finally, the generated trajectory must satisfy the physical constraints of the QCar2:

$$|\varphi(t)| \leq \varphi_{\max}, \qquad |\omega^s(t)| \leq \omega_{\max}^s.$$

If these constraints are violated, the waypoint geometry requires a turn that is too sharp, or the prescribed segment time is too short. In this case, the waypoints should be smoothed, or the corresponding segment time should be increased.